

# Peichun Hua

Computer Science and Engineering, School of Data Science  
The Chinese University of Hong Kong, Shenzhen  
[Github](#) — [Website](#) — [peichunhua@link.cuhk.edu.cn](mailto:peichunhua@link.cuhk.edu.cn)  
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## EDUCATION

**The Chinese University of Hong Kong**, Shenzhen (CUHK-Shenzhen) September 2022 — June 2026  
B.Eng in Computer Science and Engineering, School of Data Science (SDS)  
Overall GPA: 3.92/4.00 — Major GPA: 4.00/4.00 — Rank: 3/329 (School); 1/132 (Major)  
**University of North Carolina**, Chapel Hill September 2024 - May 2025  
Overall GPA: 4.00/4.00  
Exchange Student Computer Science Department

## RESEARCH INTERESTS

- Responsible AI; Adversarial Machine Learning; (ACL 2026, Work Under Review)
- AI and IoT Security; Edge Computing Security; Hardware Security; (ACSAC 2025, ISCAS 2026, SMC 2026 Under Review, TODAES Under Review)
- Differential Privacy; (Work Under Review)

## PUBLICATION

**Peichun Hua**, Danyang Chen, Diwen Xue, Mingyu Li, Yunming Xiao. “Rediscovering Deep Hashing in Secure Outsourced Dense Retrieval.” *Under Review*.

**Peichun Hua**, Yue Zheng. “ChaosFormer: Hardware-Bound Edge Transformer Model Obfuscation for Intellectual Property Protection.” *Under Review*, [ACM TODAES](#).

Xin Wang, **Peichun Hua**, Chip Hong Chang, Wenye Liu, Yue Zheng. “A Unified Open-Set Framework for Scalable PUF-Based Authentication of Heterogeneous IoT Devices.” *Under Review*, [IEEE SMC 2026](#), [Paper \(arXiv\)](#).

Danyang Chen, Yufeng Gu, Yibo Huang, Chengxuan Pei, **Peichun Hua**, Yang Zhou, Yunming Xiao. “Unlocking Software-defined GPU Fabric Scheduling in the LLM Era.” *Under Review*.

Yanchuan Tang, Junnan Li, Zhan Cheng, **Peichun Hua**, Skylar Zhai, TianMing Sha, Yuan Gao, Ruixiang Tang, Yuanyuan Lei. “MAGMA: Defending Federated Learning via Intrinsic Manifold Geometry.” *Under Review*.

**Peichun Hua**, Hao Li, Shanghao Shi, Zhiyuan Yu, Ning Zhang. “Rethinking Jailbreak Detection of Large Vision Language Models with Representational Contrastive Scoring.” To appear at [ACL 2026](#). (**Main Conference, Oral Presentation**, CORE: A\*, AR =  $\sim 2300/12148 \approx 19\%$ , [Code](#), [Paper](#)).

**Peichun Hua**, Hanxiu Zhang, Tuo Li, Yue Zheng, Wenye Liu. “Live Demonstration: Hardware-Bound IP Protection for Edge-deployed Transformers.” To Appear at [IEEE ISCAS 2026](#) (CORE: C, AR =  $1032/2012 \approx 51.3\%$ , [Paper](#)).

**Peichun Hua**, Hanxiu Zhang, Tuo Li, Yue Zheng. “Securing On-device Transformer with Hardware Binding and Reversible Obfuscation.” In [IEEE ACSAC 2025](#) (CORE: A, AR =  $84/446 \approx 18.8\%$ , [Code](#), [Paper](#)).

## AWARDS

### Academic Honors

- **Outstanding Graduate**, School of Data Science, CUHK-Shenzhen 2026
- Dean’s List, School of Data Science, CUHK-Shenzhen 2022–2023, 2023–2024
- Academic Performance Scholarship (Class B, Top <2%), School of Data Science 2023–2024
- Academic Performance Scholarship (Class C, Top <5%), School of Data Science 2022–2023
- University Research Award (with scholarship) Fall 2024, Spring 2025

### Research Awards

- *Student Conferencship*, 41st Annual Computer Security Applications Conference, [ACSAC 2025](#)
- *Artifact Available*, *Artifact Reviewed*, and *Artifact Reproducible* Badges — IEEE, for the paper “Securing On-device Transformer with Hardware Binding and Reversible Obfuscation” at [ACSAC 2025](#)
- International Student Research Internship Program (Fully Funded) McKelvey School of Engineering, Washington University in St. Louis Summer 2025

## EXPERIENCE

**School of Data Science, CUHK-Shenzhen** Shenzhen, China  
*Undergraduate Research Intern* September 2024 – May 2025

- Advisor: [Prof. Yunming Xiao](#)

- Topic: Utilizing Differential Privacy, Trusted Execution Environment, and Deep Hashing techniques to create secure outsourced dense retrieval in the RAG pipeline, with strong defense against privacy attacks while retaining performance and high throughput.
- Work completed independently. Resulted paper under review.

**McKelvey School of Engineering, Washington University in St. Louis**  
*Summer Research Intern*

Missouri, USA  
 May 2025 – August 2025

- Fully funded by [International Student Research Internship Program](#)
- Advisor: [Prof. Ning Zhang](#)
- Topic: Multi-modal Jailbreak Detection with interpretability methods instantiated with statistical tests and contrastive learning. Achieve highly effective, generalizable, and lightweight attack detection.
- Work completed independently. **Paper accepted by ACL 2026 Oral.**

**School of Science and Engineering, CUHK-Shenzhen**  
*Undergraduate Research Intern*

Shenzhen, China  
 May 2024 – May 2025

- Advisor: [Prof. Yue Zheng](#)
- Topic: Protection of the IP (intellectual property) of on-device transformer models with a specialized encryption scheme and a hardware-software co-design approach to effectively prevent unauthorized usage at the edge.
- Conferred University Research Award (URA, with scholarship). Work completed independently and directly accepted by ACSAC 2025. Live Demonstration paper accepted by IEEE ISCAS 2026 and the Extended journal version under review at ACM TODAES.

## SKILLS

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- **Coursework:** Achieved *A* grade in all listed *graduate-level* coursework  
 Cryptography      3D Computer Vision      Hardware Security and Side-Channels  
 Efficient Deep Learning      Research Topics in Computer Security      Formal Methods in Computer Security
- **Language Proficiency:** TOEFL 108 (R29+L29+S23+W27), GRE 328 (V161+Q167+3.5)

## SERVICES

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- Reviewer, IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)
- Reviewer, ACL Rolling Review